Baseline Notebook

Setup Environment

```
# DO NOT MODIFY THE CODE IN THIS CELL
!pip install -q utstd

from utstd.folders import *
from utstd.ipyrenders import *

at = AtFolder(
    course_code=36106,
    assignment="AT3",
)
at.run()

import warnings
warnings.simplefilter(action='ignore')
```

```
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.9/12.9 MB 86.9 MB/s eta 0:00:0
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 4.9/4.9 MB 107.2 MB/s eta 0:00:00
ERROR: pip's dependency resolver does not currently take into account all the
umap-learn 0.5.12 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2
hdbscan 0.8.42 requires scikit-learn>=1.6, but you have scikit-learn 1.5.2 wh
Mounted at /content/gdrive

You can now save your data files in: /content/gdrive/MyDrive/36106/assignment
```

Student Information

```
group_name = "36106-26AU-AT3-Group 24"
student_name = "Nana Ama Goldwater"
student_id = "26137455"
```

```
# Do not modify this code
print_tile(size="h1", key='group_name', value=group_name)
```

group_name

36106-26AU-AT3-Group 24

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_name', value=student_name)
```

student_name

Nana Ama Goldwater

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h1", key='student_id', value=student_id)
```

student_id

26137455

---

0. Python Packages

0.a Install Additional Packages

> If you are using additional packages, you need to install them here using the command: `! pip install <package_name>`

```
# No additional packages required. scikit-learn, pandas, numpy and altair ar
```

0.b Import Packages

```
import numpy as np
import pandas as pd
import altair as alt

from sklearn.dummy import DummyClassifier
from sklearn.metrics import (
    accuracy_score, precision_score, recall_score, f1_score,
    roc_auc_score, average_precision_score,
    confusion_matrix, classification_report,
)

# Display options
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 200)
RANDOM_STATE = 42
```

A. Assess Baseline Model

```
# DO NOT MODIFY THE CODE IN THIS CELL
# Load data
try:
  X_train = pd.read_csv(at.folder_path / 'X_train.csv')
  y_train = pd.read_csv(at.folder_path / 'y_train.csv')

  X_val = pd.read_csv(at.folder_path / 'X_val.csv')
  y_val = pd.read_csv(at.folder_path / 'y_val.csv')

  X_test = pd.read_csv(at.folder_path / 'X_test.csv')
  y_test = pd.read_csv(at.folder_path / 'y_test.csv')
except Exception as e:
  print(e)
```

A.1 Generate Predictions with Baseline Model

```
import os
import numpy as np
import pandas as pd
from sklearn.dummy import DummyClassifier

RANDOM_STATE = 42
DATA_DIR     = at.folder_path

# ── Option A: variables already in memory from earlier cells ──────────────
# If X_train / y_train etc. exist in the session, use them directly and opti
splits_in_memory = all(v in globals() for v in
                       ['X_train','X_val','X_test','y_train','y_val','y_test

if splits_in_memory:
    print("Splits found in memory → using directly from memory.")
    X_train = globals()['X_train']
    X_val   = globals()['X_val']
    X_test  = globals()['X_test']
    y_train = globals()['y_train']
    y_val   = globals()['y_val']
    y_test  = globals()['y_test']

    # Attempt to save the splits to CSV for future use, but handle OSError g
    try:
        print("  Attempting to save splits to CSV for persistence...")
        os.makedirs(str(DATA_DIR), exist_ok=True) # Ensure directory exists
        for name, obj in [('X_train', X_train), ('X_val', X_val), ('X_test',
                          ('y_train', y_train), ('y_val', y_val), ('y_test',
            obj.to_csv(DATA_DIR / f'{name}.csv', index=False)
        print("  Saved successfully.")
    except OSError as e:
        print(f"  Warning: Failed to save splits to CSV due to Google Drive
```

```
# ── Option B: load from Google Drive if not in memory ──────────────────
else: # splits_in_memory is False, so try to load from disk
    if not os.path.exists(DATA_DIR / 'X_train.csv'):
        print("Files not found. Choose one of the options below:\n")
        print("  OPTION 1 — Mount Google Drive:")
        print("    from google.colab import drive")
        print("    drive.mount('/content/drive')")
        print("    DATA_DIR = '/content/drive/MyDrive/<your-folder>/'  # ← u
        print("  OPTION 2 — Upload files directly from your machine:")
        print("    from google.colab import files")
        print("    files.upload()  # select all 6 CSVs at once\n")
        print("  OPTION 3 — Re-run all earlier pipeline cells to rebuild spl
        print("             then re-run this cell.\n")
        raise FileNotFoundError(
            "CSV files missing. Follow one of the options printed above, "
            "update DATA_DIR if needed, then re-run this cell."
        )
    else:
        print("Splits not found in memory → loading from CSV...")
        # ── Load ──────────────────────────────────
        X_train = pd.read_csv(DATA_DIR / 'X_train.csv')
        X_val   = pd.read_csv(DATA_DIR / 'X_val.csv')
        X_test  = pd.read_csv(DATA_DIR / 'X_test.csv')

        y_train = pd.read_csv(DATA_DIR / 'y_train.csv')
        y_val   = pd.read_csv(DATA_DIR / 'y_val.csv')
        y_test  = pd.read_csv(DATA_DIR / 'y_test.csv')

# ── Squeeze targets to 1-D ─────────────────────────
y_train_s = y_train.squeeze('columns') if y_train.shape[1] == 1 else y_train
y_val_s   = y_val.squeeze('columns')   if y_val.shape[1]   == 1 else y_val.i
y_test_s  = y_test.squeeze('columns')  if y_test.shape[1]  == 1 else y_test.

print("Dataset shapes")
print(f"  X_train: {X_train.shape}   y_train: {y_train_s.shape}")
print(f"  X_val  : {X_val.shape}     y_val  : {y_val_s.shape}")
print(f"  X_test : {X_test.shape}    y_test : {y_test_s.shape}")

print("\nTarget class distribution (proportion of positive class = 1)")
print(f"  train: {y_train_s.mean():.4f}  ({int(y_train_s.sum())}/{len(y_trai
)
print(f"  val  : {y_val_s.mean():.4f}  ({int(y_val_s.sum())}/{len(y_val_s)})
)
print(f"  test : {y_test_s.mean():.4f}  ({int(y_test_s.sum())}/{len(y_test_s
)

# ── Baseline models ────────────────────────────────
baseline_majority   = DummyClassifier(strategy='most_frequent')
baseline_stratified = DummyClassifier(strategy='stratified', random_state=RA

baseline_majority.fit(X_train, y_train_s)
baseline_stratified.fit(X_train, y_train_s)

preds = {
    'majority'   : {s: baseline_majority.predict(globals()[f'X_{s}'])   for
```

```
    'stratified' : {s: baseline_stratified.predict(globals()[f'X_{s}']) for
}
probas = {
    'majority'   : {s: baseline_majority.predict_proba(globals()[f'X_{s}'])[
    'stratified' : {s: baseline_stratified.predict_proba(globals()[f'X_{s}']
}

print("\nBaseline models fitted. Sample predictions on the first 10 validati
print("  majority   :", preds['majority']['val'][:10])
print("  stratified :", preds['stratified']['val'][:10])
print("  actual     :", y_val_s.iloc[:10].to_numpy())
```

```
Splits found in memory → using directly from memory.
  Attempting to save splits to CSV for persistence...
  Saved successfully.
Dataset shapes
  X_train: (11410, 18)   y_train: (11410,)
  X_val  : (2445, 18)     y_val  : (2445,)
  X_test : (2445, 18)    y_test : (2445,)

Target class distribution (proportion of positive class = 1)
  train: 0.1374  (1568/11410)
  val  : 0.1374  (336/2445)
  test : 0.1374  (336/2445)

Baseline models fitted. Sample predictions on the first 10 validation rows:
  majority   : [0 0 0 0 0 0 0 0 0 0]
  stratified : [0 1 0 0 0 0 0 1 0 0]
  actual     : [1 0 1 0 0 0 0 0 0 1]
```

A.2 Selection of Performance Metrics

> Provide some explanations on why you believe the performance metrics you chose is appropriate

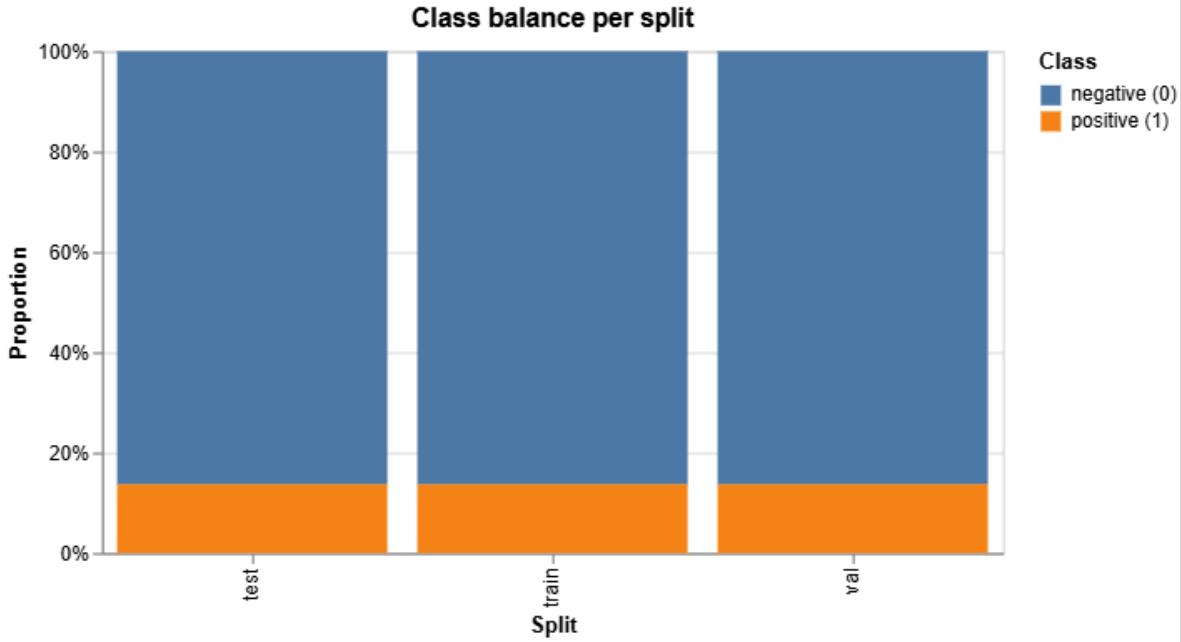
```
# Confirm class imbalance numerically — this is the main driver of metric ch
balance_df = pd.DataFrame({
    'split'   : ['train', 'val', 'test'],
    'n'       : [len(y_train_s), len(y_val_s), len(y_test_s)],
    'n_pos'   : [int(y_train_s.sum()), int(y_val_s.sum()), int(y_test_s.sum(
    'pct_pos' : [y_train_s.mean(), y_val_s.mean(), y_test_s.mean()],
})
balance_df['pct_neg'] = 1 - balance_df['pct_pos']
display(balance_df.style.format({'pct_pos': '{:.2%}', 'pct_neg': '{:.2%}'}))

# Visualise class balance per split
long_df = balance_df.melt(
    id_vars='split',
    value_vars=['pct_pos', 'pct_neg'],
    var_name='class', value_name='proportion'
).replace({'pct_pos': 'positive (1)', 'pct_neg': 'negative (0)'})

alt.Chart(long_df).mark_bar().encode(
```

```
    x=alt.X('split:N', title='Split'),
    y=alt.Y('proportion:Q', title='Proportion', axis=alt.Axis(format='%')),
    color=alt.Color('class:N', title='Class'),
    tooltip=['split', 'class', alt.Tooltip('proportion:Q', format='.2%')]
).properties(width=450, height=250, title='Class balance per split')
```

	split	n	n_pos	pct_pos	pct_neg
0	train	11410	1568	13.74%	86.26%
1	val	2445	336	13.74%	86.26%
2	test	2445	336	13.74%	86.26%




```
performance_metrics_explanations = """
Use case: repeat-purchase classificati
will place at least one order in the p
Target is binary (1 = customer reorder

Class balance is the deciding factor f
is typically imbalanced (most customer
accuracy alone is misleading a model t
while being commercially useless. The

Primary metrics
  - F1 score (positive class). Harmoni
    we actually care about (customers
    false positives (wasted marketing
  - ROC-AUC. Threshold-independent; te
    above non-reorders. Useful when dc
    (e.g. 'target the top-decile by sc
  - Average Precision (PR-AUC). On imb
    than ROC-AUC because it focuses on
    model's PR-AUC equals the positive
    interpretable in business terms.

Supporting metrics
```

```
Supporting metrics
  - Precision (positive class). What f
    to reorder' actually did? Drives c
  - Recall (positive class). What frac
    Drives coverage / opportunity cost
  - Confusion matrix. Reported on the
    (TP / FP / TN / FN) can be tied di
  - Accuracy. Reported for completenes

Reading the baselines
  - DummyClassifier(strategy='most_fre
    (the 'do nothing' strategy). Its F
    is 0.5 by construction. It establi
  - DummyClassifier(strategy='stratifi
    Its F1 ≈ positive-class rate and i
    random-ranking floor.
Any candidate model in the classificat
F1 (positive class) AND on PR-AUC to b
"""
```

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='performance_metrics_explanations', value=performa
```

performance_metrics_explanations

Use case: repeat-purchase classification predict, for each customer, whether they will place at least one order in the prediction window after the cutoff date. Target is binary (1 = customer reorders, 0 = does not). Class balance is the deciding factor for metric choice. Retail repeat-purchase data is typically imbalanced (most customers don't reorder in any given short window), so accuracy alone is misleading a model that always predicts 0 can score 80% accuracy while being commercially useless. The chosen metrics below address that. Primary metrics - F1 score (positive class). Harmonic mean of precision and recall on the class we actually care about (customers who will reorder). It is sensitive to both false positives (wasted marketing spend) and false negatives (missed customers). - ROC-AUC. Threshold-independent; tells us how well the model RANKS reorders above non-reorders. Useful when downstream teams will choose their own cutoff (e.g. 'target the top-decile by score'). - Average Precision (PR-AUC). On imbalanced data, PR-AUC is more informative than ROC-AUC because it focuses on the minority (positive)

```
ethics_metrics_note = """
Ethics note — why metric choice matters for fairness

Accuracy is rejected as a primary metric not only because it is dominated
by the majority class, but also for an ethics reason: an accuracy-
optimising model is incentivised to ignore the minority (positive) class
entirely. In a retention context that means systematically de-prioritising
customers who would have benefited from outreach likely overlapping
with under-engaged, regional, or otherwise marginal segments.

F1 and PR-AUC put the minority class at the centre of the optimisation,
which is the metric-level safeguard against silent under-service.
"""
print_tile(size="h3", key='ethics_metrics_note', value=ethics_metrics_note)
```

random-ranking floor. Any candidate model in the classification notebook MUST beat both baselines on F1 (positive class) AND on PR-AUC to be considered useful.

ethics_metrics_note

Ethics note — why metric choice matters for fairness Accuracy is rejected as a primary metric not only because it is dominated by the majority class, but also for an ethics reason: an accuracy- optimising model is incentivised to ignore the minority (positive) class entirely. In a retention context that means systematically de-prioritising customers who would have benefited from outreach likely overlapping with under-engaged, regional, or otherwise marginal segments. F1 and PR-AUC put the minority class at the centre of the optimisation, which is the metric-level safeguard against silent under-service.

A.3 Baseline Model Performance

> Provide some explanations on model performance

```
#  Scoring helper
def score_split(y_true, y_pred, y_proba):
    return {
        'accuracy' : accuracy_score(y_true, y_pred),
        'precision': precision_score(y_true, y_pred, zero_division=0),
        'recall'   : recall_score(y_true, y_pred, zero_division=0),
        'f1'       : f1_score(y_true, y_pred, zero_division=0),
        'roc_auc'  : roc_auc_score(y_true, y_proba) if len(np.unique(y_true)
        'pr_auc'   : average_precision_score(y_true, y_proba) if len(np.uniq
    }

y_by_split = {'train': y_train_s, 'val': y_val_s, 'test': y_test_s}
rows = []
for model_name in ['majority', 'stratified']:
    for split in ['train', 'val', 'test']:
        m = score_split(y_by_split[split], preds[model_name][split], probas[
        m['model'] = model_name
        m['split'] = split
        rows.append(m)

results = pd.DataFrame(rows)[
    ['model', 'split', 'accuracy', 'precision', 'recall', 'f1', 'roc_auc', '
]
print("Baseline performance — all splits, both strategies:")
display(
    results.style.format({c: '{:.4f}' for c in
```

```
        ['accuracy', 'precision', 'recall', 'f1', 'roc_auc', 'pr_auc']})
)

# Confusion matrix on validation (majority strategy)
cm = confusion_matrix(y_val_s, preds['majority']['val'])
print("\nConfusion matrix — majority baseline on validation set:")
print(pd.DataFrame(
    cm,
    index=['actual_0', 'actual_1'],
    columns=['pred_0', 'pred_1']
))

print("\nClassification report — stratified baseline on validation set:")
print(classification_report(y_val_s, preds['stratified']['val'], zero_divisi

# Visualise metric comparison on validation
val_long = (
    results.query("split == 'val'")
           .melt(id_vars=['model', 'split'],
                 value_vars=['accuracy', 'precision', 'recall', 'f1', 'roc_a
                 var_name='metric', value_name='score')
)
alt.Chart(val_long).mark_bar().encode(
    x=alt.X('model:N', title=None, axis=alt.Axis(labels=False, ticks=False)),
    y=alt.Y('score:Q', title='Score', scale=alt.Scale(domain=[0, 1])),
    color=alt.Color('model:N', title='Baseline'),
    column=alt.Column('metric:N', title='Metric',
                      sort=['accuracy', 'precision', 'recall', 'f1', 'roc_au
    tooltip=['model', 'metric', alt.Tooltip('score:Q', format='.4f')]
).properties(width=80, height=220, title='Baseline metrics on the validation
```

```
Baseline performance — all splits, both strategies:
```

	model	split	accuracy	precision	recall	f1	roc_auc	pr_auc
0	majority	train	0.8626	0.0000	0.0000	0.0000	0.5000	0.1374
1	majority	val	0.8626	0.0000	0.0000	0.0000	0.5000	0.1374
2	majority	test	0.8626	0.0000	0.0000	0.0000	0.5000	0.1374
3	stratified	train	0.7635	0.1322	0.1295	0.1308	0.4970	0.1367
4	stratified	val	0.7550	0.1188	0.1220	0.1204	0.4889	0.1352
5	stratified	test	0.7607	0.1391	0.1429	0.1410	0.5010	0.1377

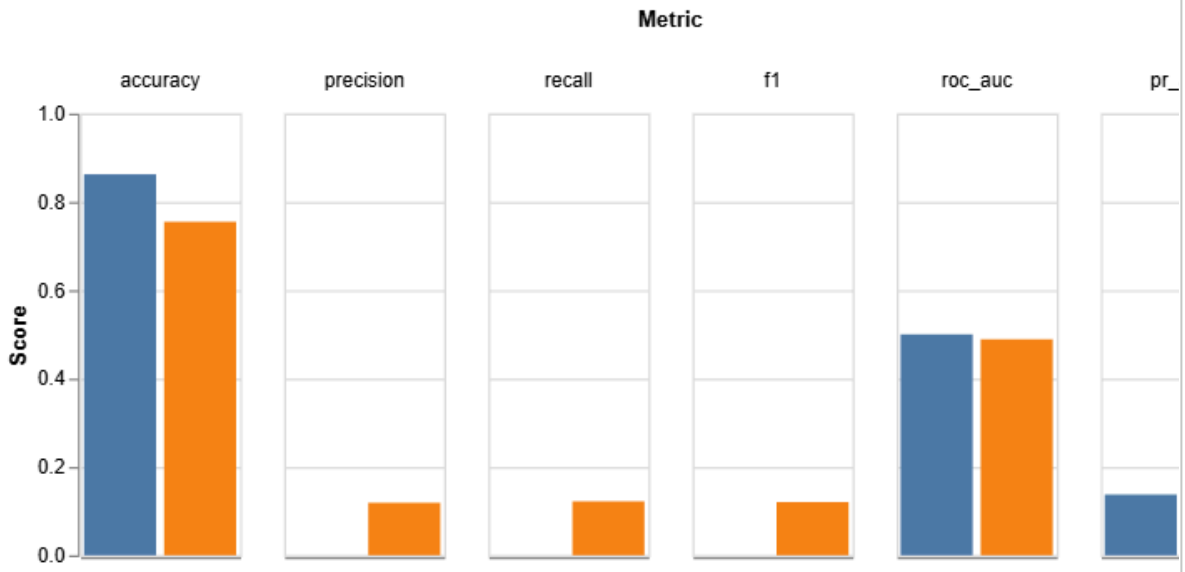
```
Confusion matrix — majority baseline on validation set:
          pred_0  pred_1
actual_0    2109       0
actual_1     336       0

Classification report — stratified baseline on validation set:
              precision    recall  f1-score   support

           0       0.86      0.86      0.86      2109
           1       0.12      0.12      0.12       336

    accuracy                           0.76      2445
   macro avg       0.49      0.49      0.49      2445
weighted avg       0.76      0.76      0.76      2445
```

Baseline metrics on the validation set




```
baseline_performance_explanations = ""
Headline numbers (validation set)
  - majority   : accuracy ≈ proportion
                 recall, F1 on the pos
                 and PR-AUC are at the
  - stratified : accuracy drops toward
                 positive rate); F1 ≈
  - Train vs val vs test scores are es
    train and val for a real model is
```

```
    to dataset noise.

Interpretation
  - The majority baseline highlights w
    it scores high on accuracy while b
    a single customer as 'will reorder
    but not on F1 or PR-AUC has learne
  - The stratified baseline provides a
    must clear to demonstrate genuine
    positive-class rate, which is the
    PR-AUC on imbalanced data.
  - The confusion matrix for the major
    From a business standpoint this me
    (recall = 0). This is the cost of
    business floor any deployed model

Acceptance criteria for the classifica
  - F1 on positive class > stratified
  - PR-AUC > positive-class rate (i.e.
  - ROC-AUC meaningfully above 0.5 (ta
  - Test-set performance within ~1–2 p
    gaps suggest validation leakage or
  - Confusion-matrix-derived precision
    business terms (campaign efficienc
"""
```

```
# DO NOT MODIFY THE CODE IN THIS CELL
print_tile(size="h3", key='baseline_pe
```

baseline_performance_explanations

Headline numbers (validation set) - majority : accuracy ≈ proportion of the majority (negative) class; precision, recall, F1 on the positive class are all 0